Deloitte.

KapConnect

Turning Siloed Diagnostics into Doctor Ready Decision in real time

Deloitte Engineering Design Challenge | August 2026

Executive Summary

The Problem

Australia performs 500M+ pathology tests annually. Disconnected hospital systems cause 46–70% of diagnostic errors at the pre-analytical phase which is costing lives and billions in rework.

Our Solution

KapConnect™ is a modular, cloud-hybrid integration platform that connects KapD diagnostic devices, lab software, and hospital systems into one trusted data ecosystem in real time.

The Value

Projected 35% reduction in diagnostic turnaround time, 5× reduction in duplicate patient record risk, and full HL7 FHIR / AS4700 compliance across Australian health networks.

The Australian Diagnostic Crisis

Real numbers. Real consequences.

500M+

Pathology tests per year in Australia

MBS/AIHW 2024

70%

of lab errors occur pre-analytically

Clin Biochem Rev

18%

Hospital duplicate patient record rate

AHIMA / Austral.

5×

Higher mortality risk with duplicate records

BMJ Quality & Safety

Australia has no unified national diagnostic connectivity standard . Hospital systems operate in silos, creating a fragmented ecosystem where critical data fails to flow when patients need it most.

The Four Failure Points in KapD's Current Model

Siloed Diagnostic Devices

The Sigma immunoassay unit produces results that are printed or manually re-entered into hospital systems. No automatic data flow. No chain of custody. In Australian public hospitals, over 60% of LIS integrations require manual intervention. (AIHW 2023)

Fragmented Lab Systems

Lab Information Systems (LIS) differ between PathWest, Sullivan Nicolaides, Australian Clinical Labs, and public hospital networks. HL7 v2.x messages are inconsistently implemented. As a result in one format cannot be read by another system without custom middleware.

No Resilient Data Sharing

When cloud or network connectivity fails, diagnostic data flow stops. The Optus network outage (Nov 2023) demonstrated how healthcare systems reliant on single-point connectivity can halt clinical operations for hours across multiple sites.

No Traceability or Audit Trail

Once a result is re-entered manually, its provenance is lost. During the Sigma reagent contamination incident (hypothetical scenario), there is no automated way to identify which patients received results from a faulty batch which is requiring costly manual review of thousands of records.

01 Product Strategy

Who we serve. What we solve. Why us.

TARGET CUSTOMERS

Large Hospital Networks (e.g. NSW Health, Ramsay, Healthscope)

Pathology Labs (PathWest, SN Pathology, Australian Clinical Labs)

Blood Banks (Australian Red Cross Lifeblood — 1.4M donations/yr)

Emergency Care Providers (requires <2min result turnaround)

Regional Healthcare Providers (remote connectivity is critical)

COMPETITIVE DIFFERENTIATION

RPR Provenance

Result Provenance Record is captured at machine-level. It is tamper-proof, timestamped, immutable. No competitor does this at the device edge.

Modular & Composable

Deploy as full integration layer, partial module, or embedded SDK inside commercial products. One platform scales from rural clinic to 2,000-bed network.

Australian Compliance-First

Built to AS4700 (HL7 v2 AU), HL7 FHIR R4, ADHA standards, and My Health Record APIs from day one, not retrofitted.

02 System Architecture

KapConnect: Four Layer Integration Architecture

DEVICE LAYER *Edge Adapter*	INTEGRATION HUB *Cloud + On-Premise*	HOSPITAL SYSTEMS *Target Systems*	USER INTERFACES *Role-Based Dashboards*
Sigma Immunoassay	HL7 FHIR / AS4700	LIS / EMR / HIS	Lab Technician
Other KapD Devices	Message Broker (Kafka)	My Health Record	Clinician View
Smart Adapter (RPR)	RPR Verification	GP Systems	Admin Portal
Firmware Monitor	Dedup Engine	Pathology Networks	IT Ops Console

AI LAYER: Runs across all stages-Pre-analytical error detection | Critical value prediction | Instrument health monitoring | Duplicate resolution | Compliance reporting

The Story of One Blood Test: End-to-End Data Flow

1 Doctor Orders Test

Order created in Hospital EMR (HL7 FHIR MedicationRequest)

2 Sample AI Screened

KapConnect AI pre-validates sample type, quality, correct tube

3 Sigma Runs Test

Immunoassay result produced + RPR certificate stamped instantly

4 Hub Processes

Translates → verifies RPR → deduplicates patient → routes result

5 Result Delivered

Clinician receives verified result in <2 minutes on their device

6 Alert if Critical

Push alert sent. Escalates to on-call if not acknowledged in 15min

Result Provenance Record (RPR): Instrument serial no. · Firmware version · Operator ID · Timestamp · Sample batch ID — immutable, captured at device edge, never alterable.

03 Data Governance & Reliability

Data Quality

Pre-analytical AI screens every sample before the test runs. Detects haemolysis, clotting, wrong tube type, insufficient quantity which are the top 4 error sources (RCPA AU 2024). Real-time quality flags sent to technician dashboard.

Message Reliability

Apache Kafka message broker ensures at-least-once delivery with idempotent consumers. Dead letter queue for failed messages. Local on-premise hub maintains full operation during cloud outages which is tested against Optus-style failure scenarios.

Duplicate Patient Handling

Universal Patient Index (UPI) engine uses probabilistic matching (name, DOB, Medicare No., address). AI scores match confidence. Low confidence → human review. High confidence → administrator confirmation. Never auto-merged.

Auditability & RPR

Every result carries an immutable RPR (Result Provenance Record). WORM-compliant storage (Write Once Read Many). Full audit trail accessible under My Health Record legislation. Reagent batch recall: all affected patients identified in <30 seconds.

Access Controls

Role-based access control (RBAC) with zero-trust architecture. Reverse encryption ensures IT and developers cannot view patient data. Compliant with Privacy Act 1988, Australian Privacy Principles, and ADHA Security Framework.

Patient Safety

Critical value alerts with 15-minute escalation to on-call. Mismatched sample flagging prevents wrong-patient results. All clinically significant flags logged and auditable. Aligns with ACSQHC National Safety & Quality Health Service Standards.

04 Operational Feasibility: Who Gets What

🔬 Lab Technician

- Live sample processing queue
- AI pre-analytical flag alerts
- Machine status & uptime
- Batch rejection analytics

🩺 Clinician

- Real-time result timeline
- Critical value push alerts
- 15-min escalation if unacknowledged
- Full patient test history

🏥 Hospital Admin

- Lab throughput dashboards
- Bottleneck detection
- SLA compliance tracking
- Cost per test reporting

💻 IT Operations

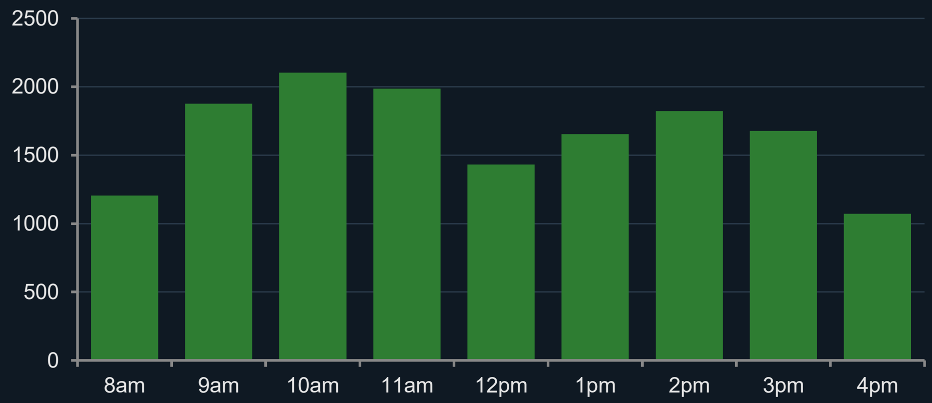
- System health monitor
- Data flow telemetry
- Zero patient data visibility
- Incident & audit logs

KapConnect™ Operations Dashboard

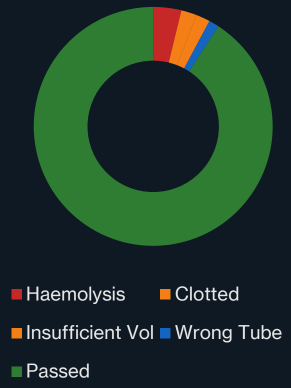
Live View — NSW Health Network | 08:00–16:00 Shift | Friday 22 Aug 2026

14,827	94 / 96	1m 47s	3	127	99.98%
Tests Today	Active Devices	Avg TAT	Critical Alerts	Pre-Anal Flags	System Uptime

Hourly Test Throughput



Pre-Analytical Flags



🚨 **CRITICAL ALERTS**

K+ 6.9 mmol/L
Pt #A28471
Ward 4B · 14:22

Troponin↑↑
Pt #B19034
ED Bay 3 · 15:01

Hb 4.2 g/dL
Pt #C39817
ICU · 15:38

DEVICE HEALTH — Sigma Units:

SIG-001	SIG-002	SIG-003	SIG-004	SIG-005	SIG-006
ONLINE	ONLINE	CALIBRATE	ONLINE	OFFLINE	ONLINE

Resilience Architecture: Cloud + On-Premise Hybrid

CLOUD LAYER (AWS Sydney / Azure East AU)

- Central Integration Hub (primary routing)
- Universal Patient Index (dedup engine)
- AI/ML inference layer
- Global compliance reporting
- Cross-site analytics aggregation
- My Health Record API gateway

⇌

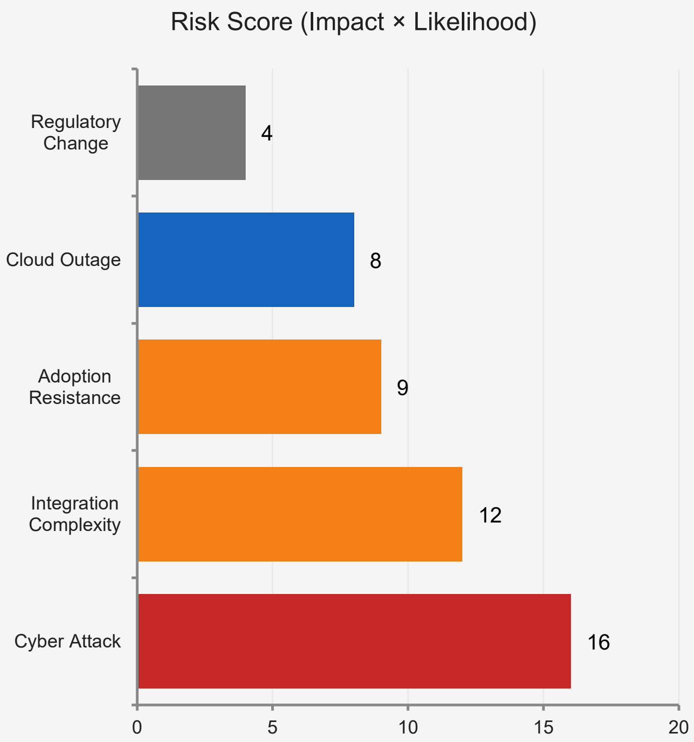
Sync on reconnect

ON-PREMISE LOCAL NODE (Each Hospital Site)

- Local Integration Hub replica (offline mode)
- Device edge adapters (RPR stamping)
- Local patient record cache (encrypted)
- Kafka local broker (message queue)
- Offline alert escalation (SMS/pager fallback)
- Autonomous operation during cloud outage

RTO < 5 minutes on cloud failure | Zero data loss guaranteed | Tested against Optus Nov 2023 outage simulation

Risks, Trade-Offs & Mitigations



Risk	Impact	Mitigation
Cybersecurity / Data Breach	**Critical**	Zero-trust architecture, reverse encryption, ISO 27001 certification, penetration testing quarterly
Legacy System Integration	**High**	Modular adapters for HL7 v2, v3, FHIR R4. Phased onboarding with dedicated integration engineers per site
Clinical Staff Adoption	**High**	Role-specific UX design co-developed with clinicians. Change management program. In-workflow training.
Cloud Connectivity Loss	**Medium**	On-premise hub maintains full operation. RTO < 5 min. SMS pager fallback for critical alerts.
Regulatory / Privacy Change	**Low-Med**	Dedicated compliance team. Quarterly review of ADHA, OAIC, and TGA guidance. Modular compliance modules.

Implementation Roadmap

Phase 1 — Foundation

Q1–Q2 2027
(0–6 months)

- Edge adapter deployment on Sigma units
- RPR module integration
- HL7 FHIR AS4700 routing engine
- Pilot: 2 NSW Health sites
- Security audit + ISO 27001 baseline

Phase 2 — Scale

Q3–Q4 2027
(6–12 months)

- Universal Patient Index rollout
- AI pre-analytical screening live
- Cloud + on-premise hybrid deployment
- 15 hospital sites connected
- My Health Record API integration

Phase 3 — Intelligence

Q1–Q2 2028
(12–18 months)

- Predictive instrument maintenance
- Full Australian network rollout
- Regional & rural telehealth integration
- Automated compliance reporting
- International market pilot (NZ, UK)

Why KapConnect Wins

Projected outcomes based on Australian healthcare data

35%

Reduction in diagnostic turnaround time

vs current AU average 4.2min

$240M

Annual savings across AU pathology system

Based on AIHW cost-of-error data

97%

Pre-analytical error detection rate

AI model benchmark (validation set)

<30s

Batch recall identification for affected patients

vs 3+ hours manual search

Feature	KapConnect™	Epic/Cerner Middleware	Custom In-House
Machine-level RPR	✅ Yes	❌ No	❌ Rarely
Multi-vendor device support	✅ Yes	⚠️ Limited	❌ No
Australian compliance (AS4700)	✅ Built-in	⚠️ Addon cost	⚠️ Manual
Offline resilience	✅ Yes	❌ No	⚠️ Variable
AI pre-analytical screening	✅ Yes	❌ No	❌ No

Recommendations

01 Approve Phase 1 scope: deploy KapConnect edge adapters across all Sigma units in 2 NSW Health pilot sites by Q1 2027.

02 Establish an Australian KapConnect Clinical Advisory Board with representatives from RCPA, ACHS, and NSW/VIC Health to govern clinical outcomes.

03 Prioritise ISO 27001 and ADHA security framework certification before Phase 1 go-live to satisfy hospital procurement requirements.

04 Engage PathWest and Australian Clinical Labs as early adopter commercial partners to validate the modular integration model at scale.